HUAN CHANG

INDUSTRIAL DESIGN

PORTFOLIO 2026

Hi, I'm Huan.

I'm an emerging industrial designer and a parent based in the San Francisco Bay Area. I see design as a way to protect people's limited resources—time, energy, attention, confidence, and emotional capacity. Caring for a family has made me attentive to products that support people under real-life constraints, gently encourage better habits, and create small moments of relief, clarity, or happiness.

After exploring other paths, discovering industrial design brought me back to the joy of learning and making. The patience, focus, and care I bring to product development are the same qualities I find in crochet, miniature building, and Chinese calligraphy.

EXPERIENCE

Biodesign Finalist
Biodesign Challenge/SFSU Team
2025

Marketing & Research Assistant
BorderX Lab/Beyond App
2018-2019

Research Assistant
UI Chicago/Political Science
2012-2015

SKILLS

Digital
Rhino
SolidWorks
KeyShot
Photoshop
Illustrator
InDesign
Procreate
Figma

Physical
Sketching
3D printing
Prototyping
Laser Cutting

Language
Mandarin Chinese - Native
English - Proficient

EDUCATION

B.S. Industrial Design
San Francisco State University
2024-Present

Offsite Pro
AdvancedDesign
2025-2026

M.A. Political Science
University of Illinois at Chicago

B.A. English Language
Northwest University (China)

INTERESTS

Crochet
Embroidery
Miniature houses
Chinese calligraphy

CONTACT

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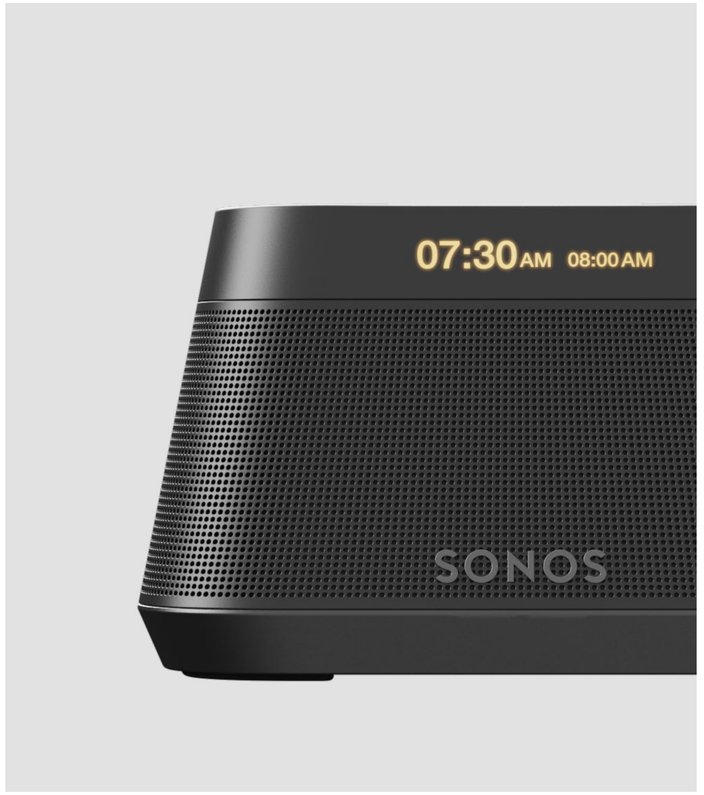
San Francisco Bay Area, CA

PROJECTS

ARIA Inhaler Spacer



WAKE Portable Spacer

PULLAID Bandage Dispenser

ARIA

Inhaler Spacer

Aria is a portable inhaler spacer that integrates the inhaler, spacer, and mask into one compact, protected system, making asthma care easier for kids and parents on the go.

Timeline: 5 weeks, 2026

CHALLENGE

6 million U.S. children live with **asthma**, with **exercise triggering** symptoms in to 40–90% of cases. Studies show that **spacers** improve medication delivery by 2-4x.

Pediatricians recommend spacers for better medication delivery. Yet they are often left behind.

cdc.gov | Klain, A., et al. (2024). Exercise-induced bronchoconstriction, allergy and sports in children. Italian Journal of Pediatrics, 50, 47 | https://www.aaaai.org/tools-for-the-public/conditions-library/asthma/spacers-and-valved-holding-chambers-(vhcs)-for-use

WHY SPACERS ARE LEFT BEHIND?

BULKY

Bulky spacers are often left behind, highlighting the need for a compact, everyday carry solution.

DISORGANIZED

Separate components make packing, finding, and setup difficult, highlighting the need for integrated storage.

CONTAMINATED

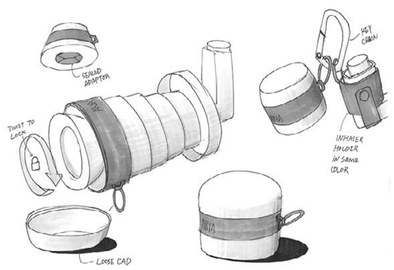
Exposed masks are easily contaminated, highlighting the need for protective storage.

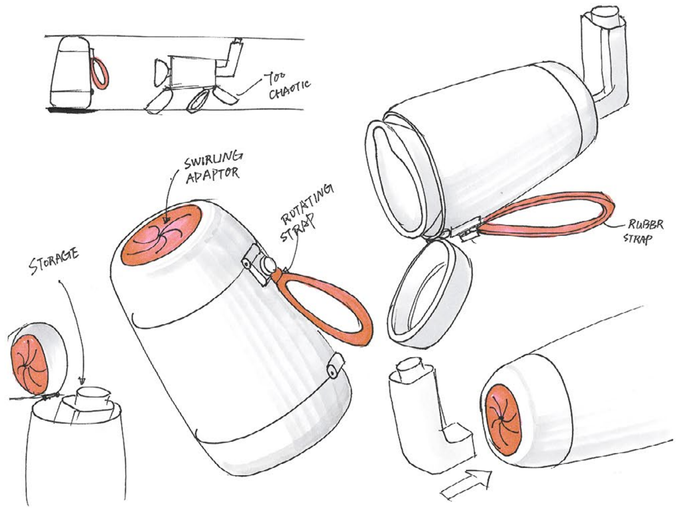
How might we encourage everyday spacer use by improving **portability**, **storage**, and **hygiene**?

IDEATION

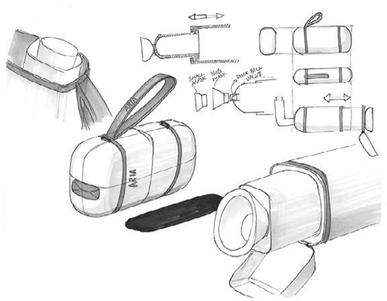
Word lists, thumbnails, and rapid concept sketches translate key challenges into potential solutions for **portability**, **storage**, and **hygiene**.

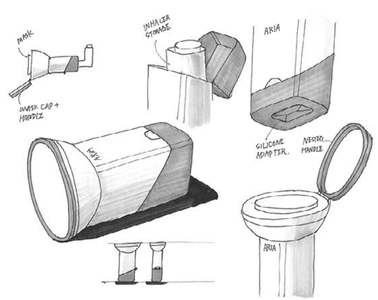
01 CONE

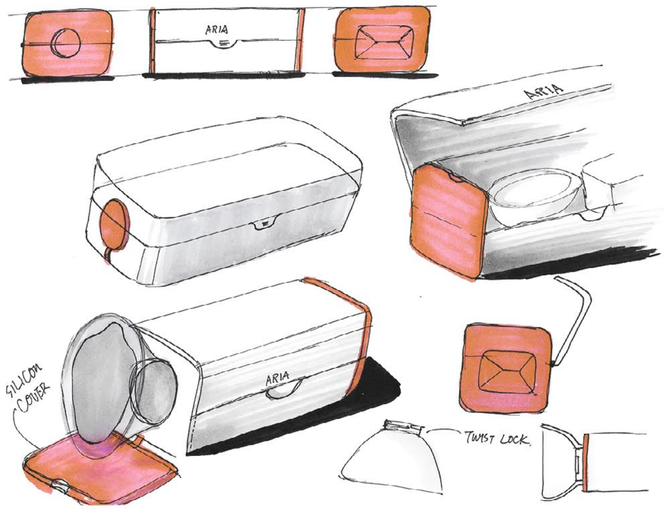




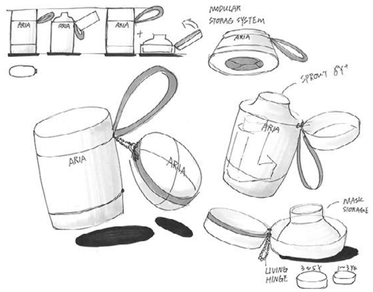
02 BOX

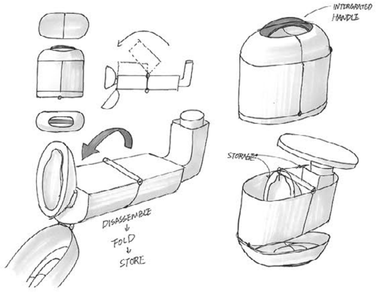


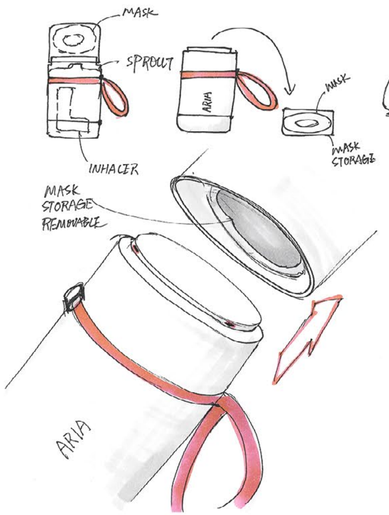


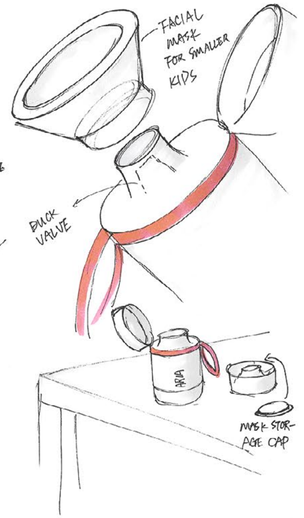


03 MODULAR







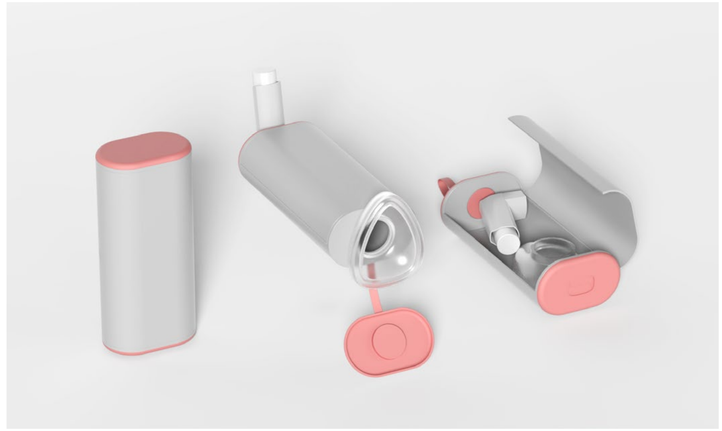


CONCEPT DEVELOPMENT

01 CONE

Quick to deploy with a simple opening mechanism, but feels **bulky** for smaller hands. The inhaler can also shift around during storage.

02 BOX



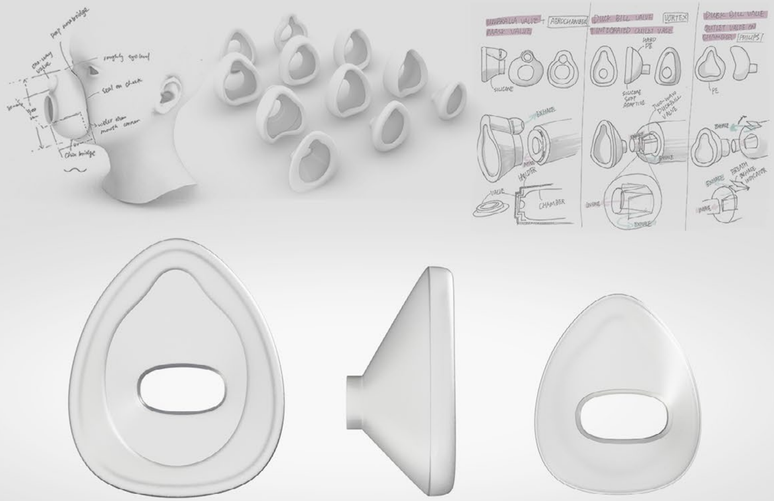
Compact storage keeps the mask and inhaler securely nested inside the chamber. But the **elongated form** and less intuitive setup make deployment more cumbersome.

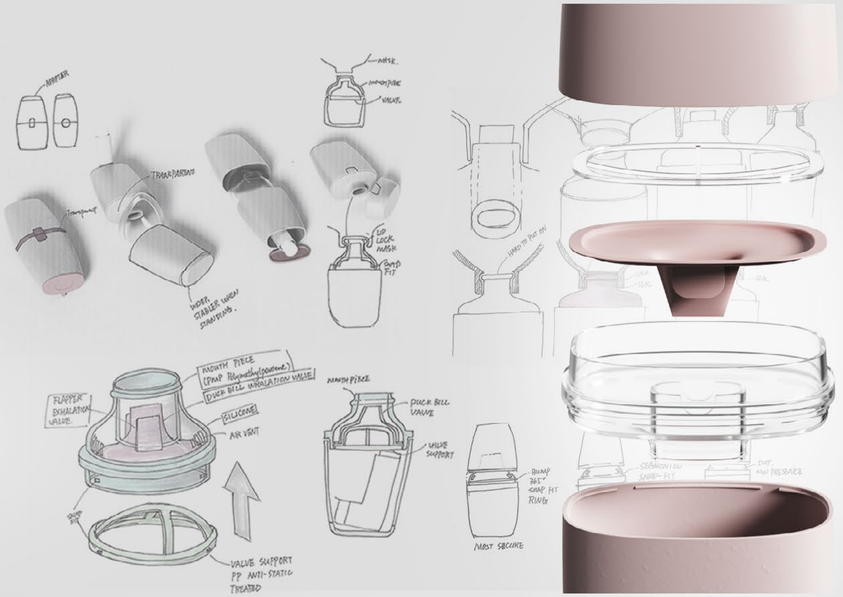
03 MODULAR

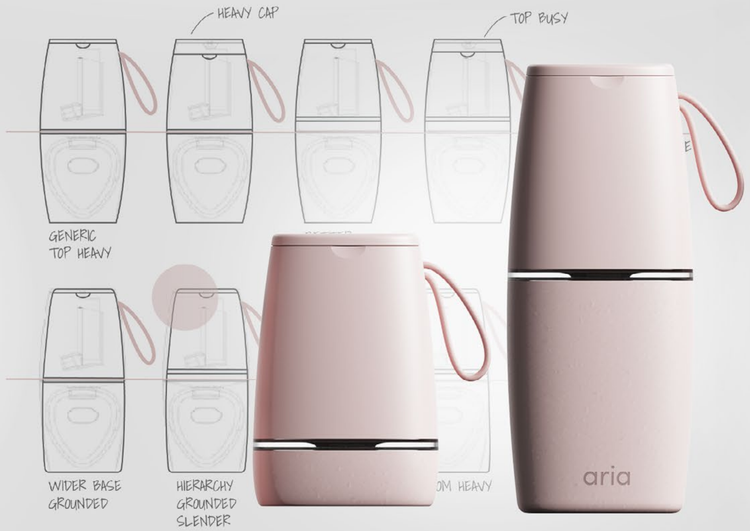
Compact and **comfortable** for small hands, with dedicated mask and inhaler storage, and a mouthpiece that **transitions easily** to a mouthpiece as kids grow.

REFINEMENT

Refinement explored **attachment and assembly strategies** alongside **mask**, **strap**, **form**, and **adapter** development to create a more cohesive and functional system.

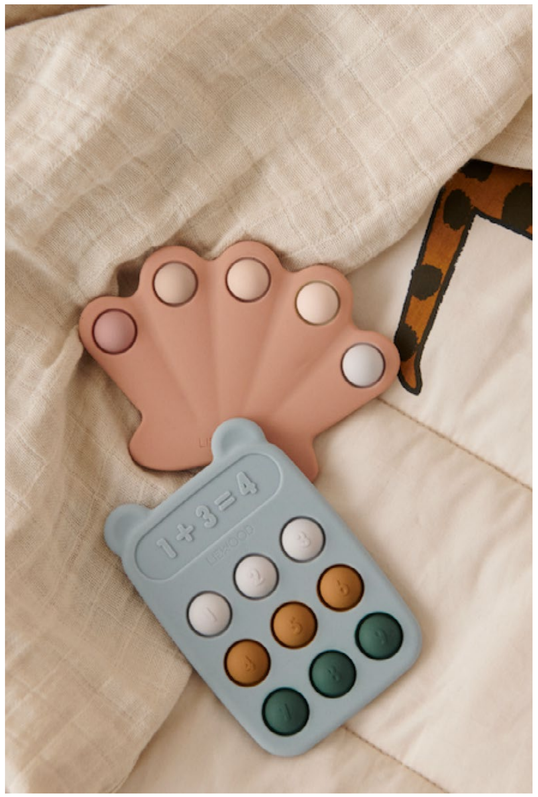


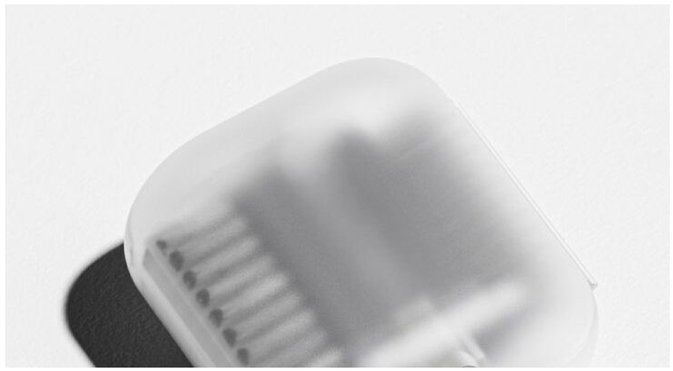




CALM LIFESTYLE

A calm lifestyle direction that balances warmth and approachability with clean, confident forms, shifting the spacer from a medical device toward a **familiar everyday carry product**.





aria

Breath freely. Play fully.

Inhaler Adaptor
Air Chamber
Valve Retainer
One-way Valve
Mouthpiece
Mask Storage

S · 0-18 mo

M · 18 mo-3 yr

L · 3-6 yr

Youth · 6+ yr

WAKE

Audio Alarm

WAKE is a sound-first bedside alarm that combines Sonos' rich premium audio with intuitive, eye-free controls, creating a calmer, more natural wake-up experience.

Timeline: 5 weeks, 2026

WHY SOUND?

Wake-up happens in a **low-attention state** where **sound is the primary wake-up cue.** Yet today's bedside products emphasize functionality, wellness, and smart features over the audio experience itself.

LOW-ATTENTION

SOUND-FIRST

SONOS

How might Sonos bring a **sound-first perspective** to the wake-up experience?

SONOS DESIGN LANGUAGE

COMPACT FORM

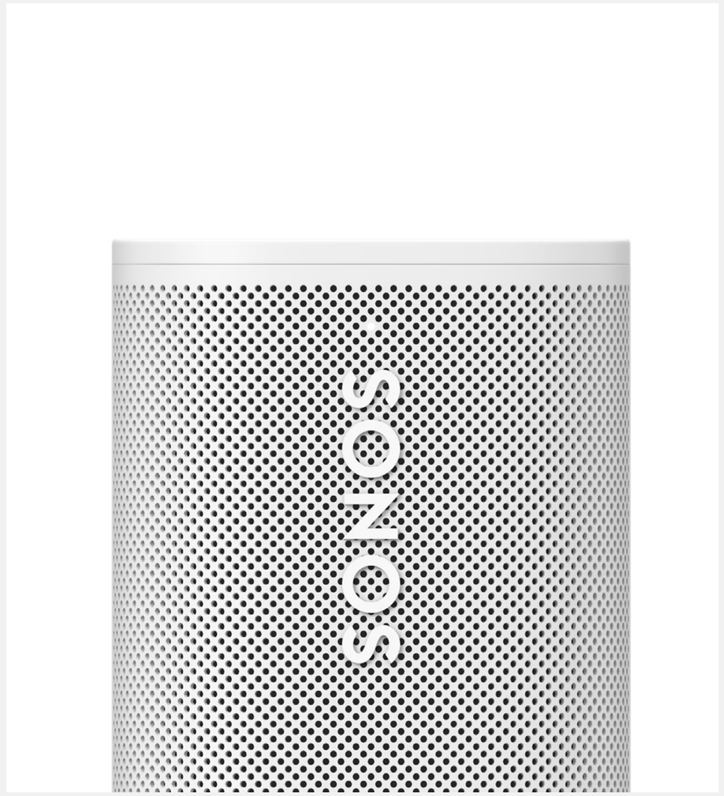
A tapered lower body visually lifts the product from the surface, creating a lighter presence while reflecting Sonos' use of softened, controlled geometry.

EFFORTLESS INTERACTION

A gently concaved top surface guides the hand toward the controls, while embossed buttons provide tactile landmarks for confident, eye-free use.

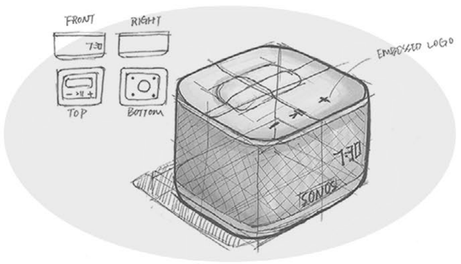
UNDERSTATED CMF

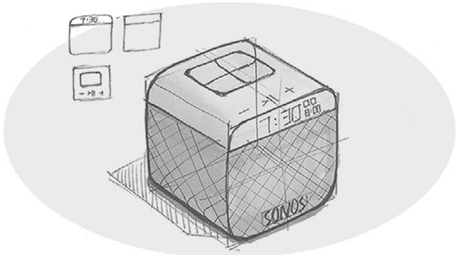
Monochrome finishes, subtle branding, and precision perforation create a refined presence that feels premium without demanding attention.

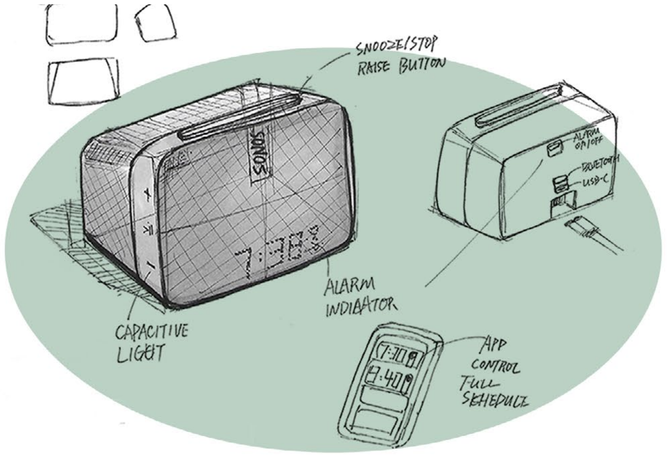


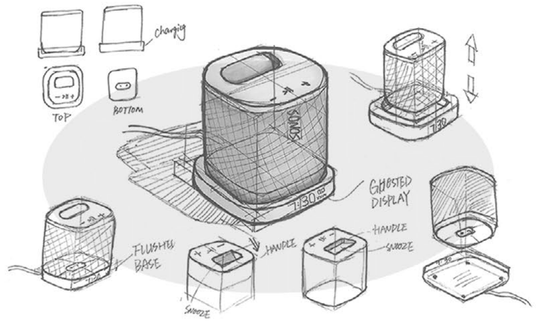
IDEATION

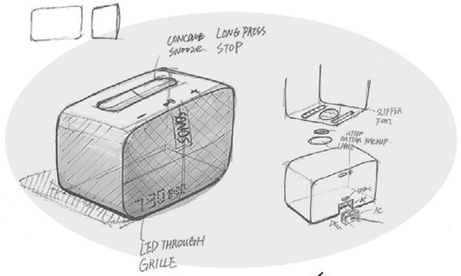
Ideation explored form factors and feature arrangements that prioritize **premium audio**, intuitive **eye-free interaction**, and a **calm** bedside presence.

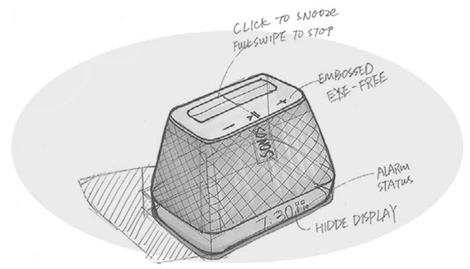


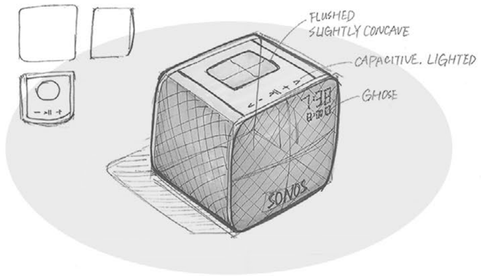


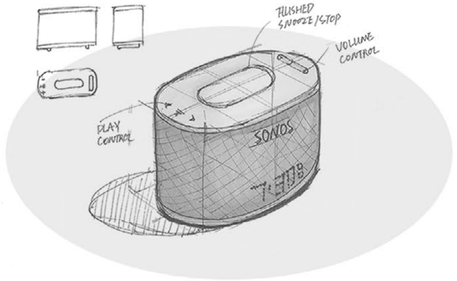


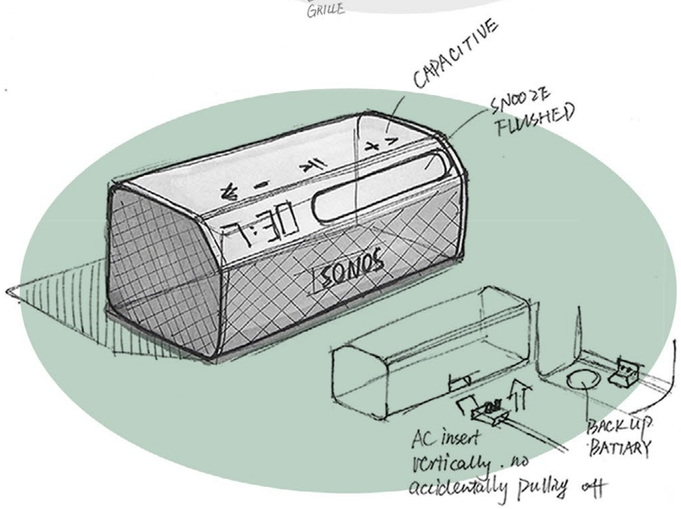


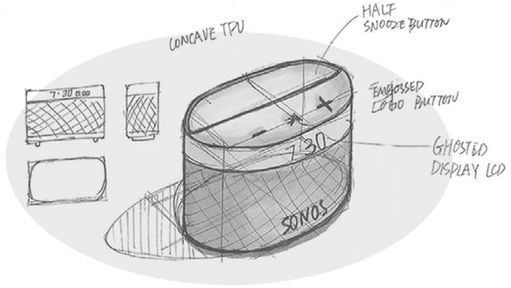


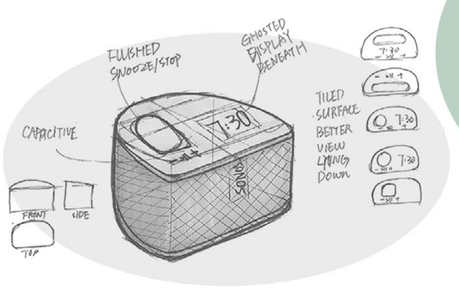


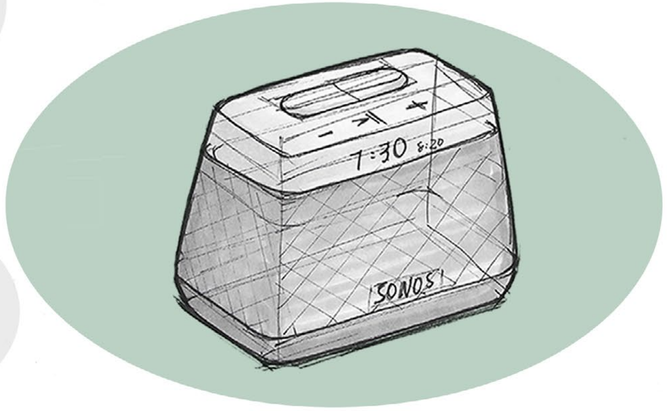












CONCEPT DEVELOPMENT

01: BAR

Pro
- Familiar & readable
- Stable acoustic form

Con
- Large footprint
- Display-dominated presence

02: POD

Pro
- Compact
- Reduced visual distraction

Con
- Hidden top control
- Interrupted front surface

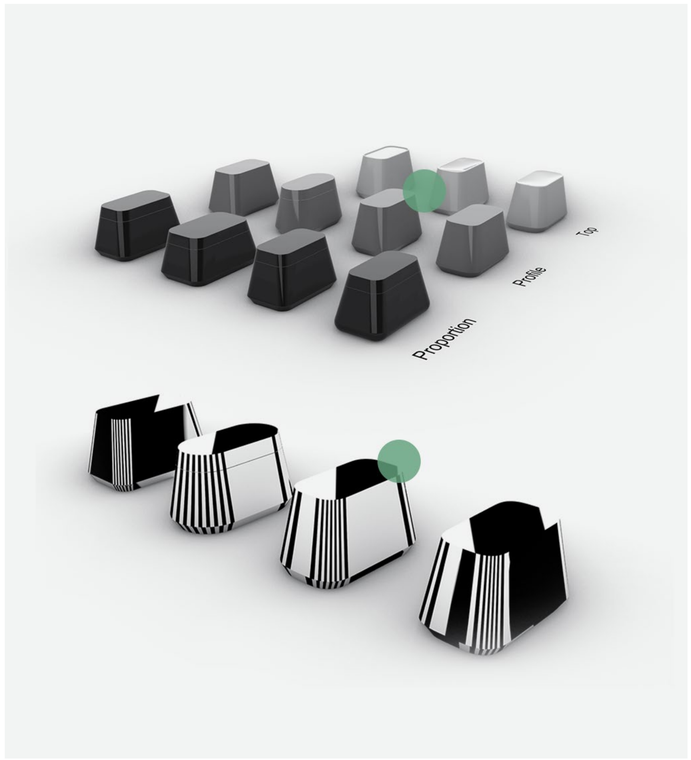
03: WEDGE

Pro
- **Audio-first expression**
- **Tactile, eye-free interaction**

Con
- Smaller display

REFINEMENT

FORM



Refined **proportions** and **curvature** for smooth reflections with defined edges.

ARCHITECTURE

Refined assembly logic around **performance-level audio** components.

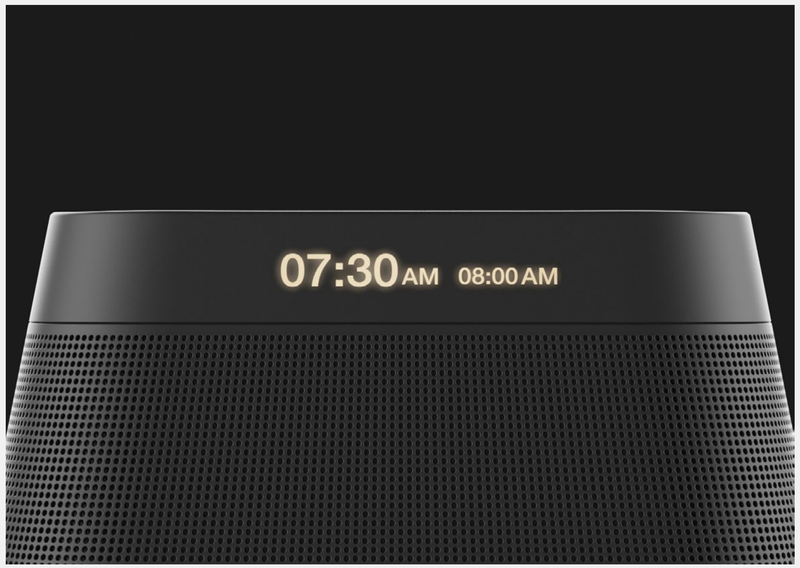
CONTROL

Explored tactile, intuitive controls for confident **eye-free interaction**.

WAKE

Premium audio meets bedside simplicity

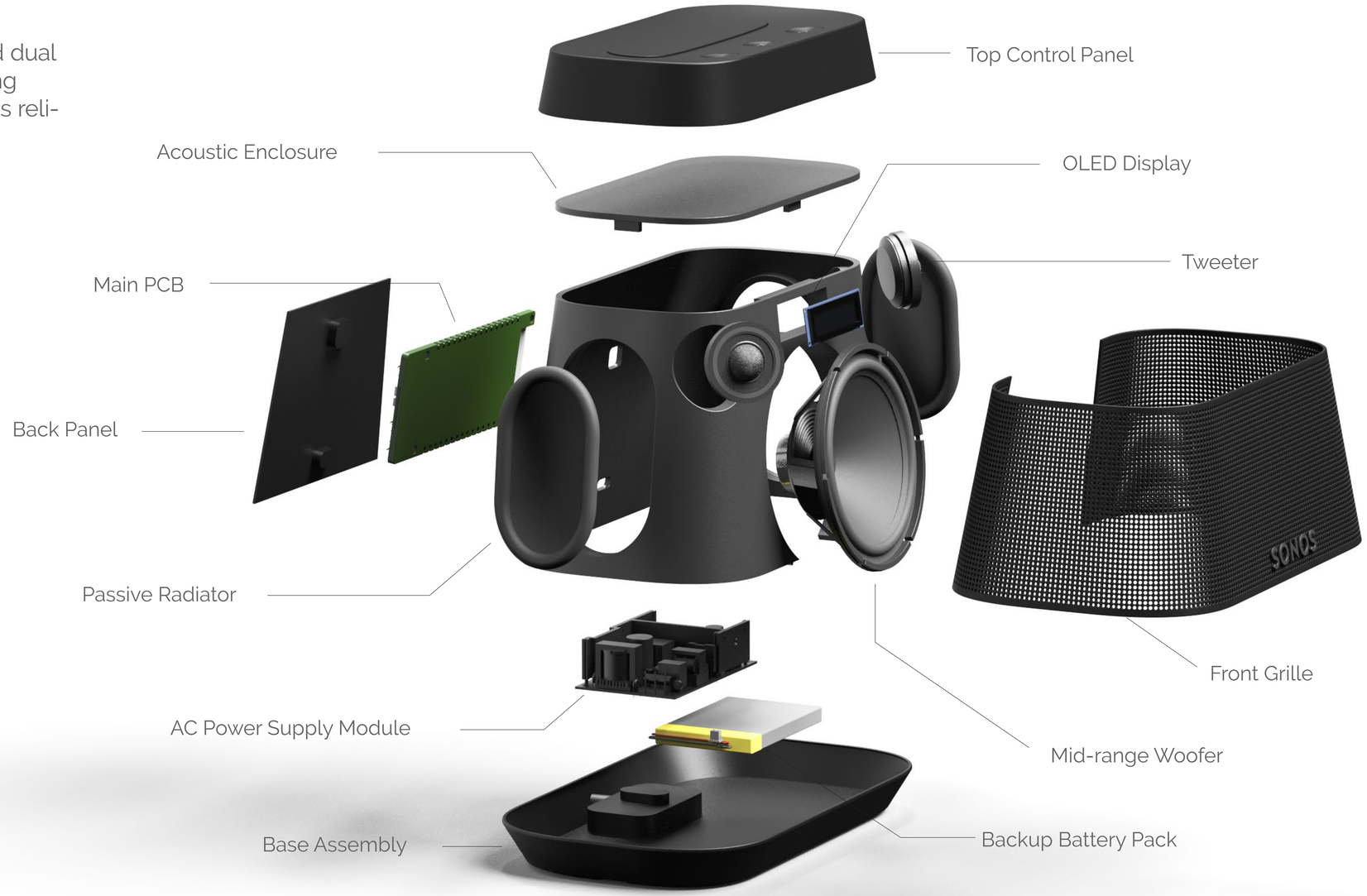
07:30AM 08:00AM
SONOS
Sonos Wake


07:30 AM 08:00 AM

SONOS
Sonos Wake

BUILD AROUND SOUND

A mid-range woofer, dual tweeters, and dual passive radiators deliver rich, room-filling sound, while a backup battery maintains reliable operation during power outages.



07:30 AM 08:00 AM
SONOS
NATHAN WILLIAMS
THE KINFOLK HOME

PullAid

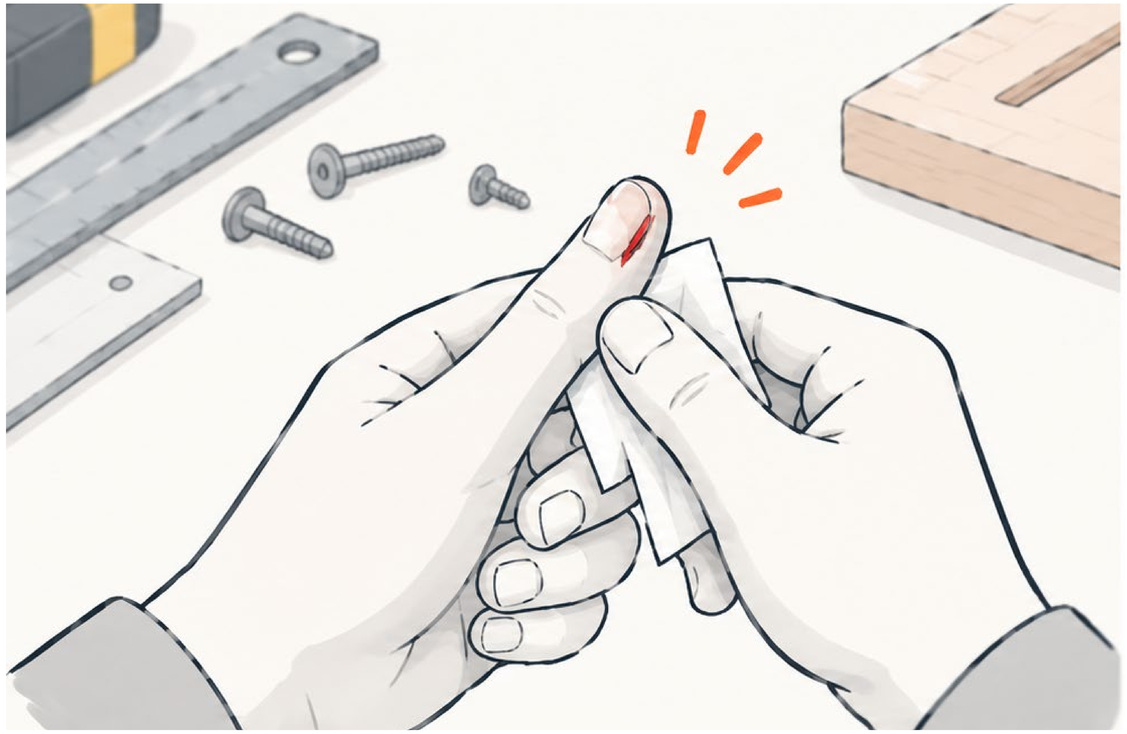
Bandage Dispenser

PullAid is a one-handed bandage dispenser that simplifies dispensing and opening, making minor wound care easier when one hand is injured or occupied.

Timeline: 6 weeks, 2026

CHALLENGE

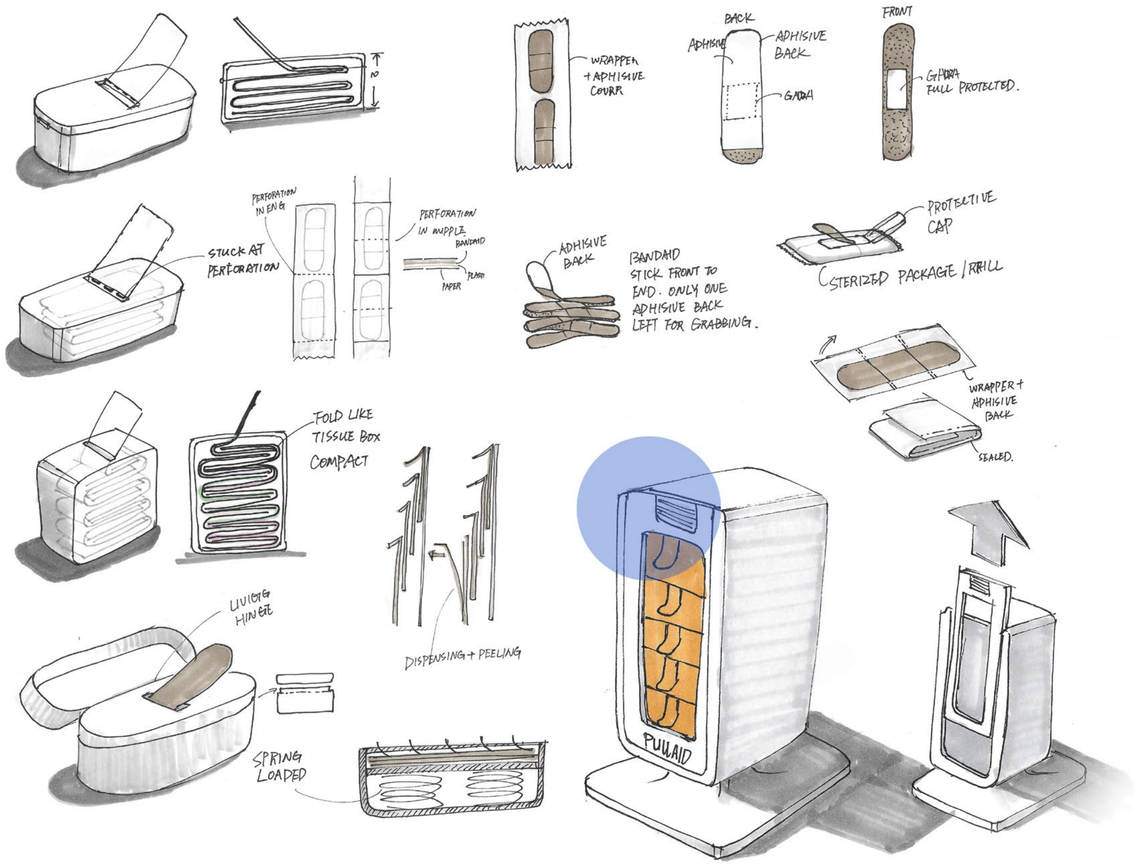
Minor cuts often leave one hand injured or occupied, yet getting a bandage ready **requires both hands**—from taking it out of the box to opening the wrapper and removing the liners.

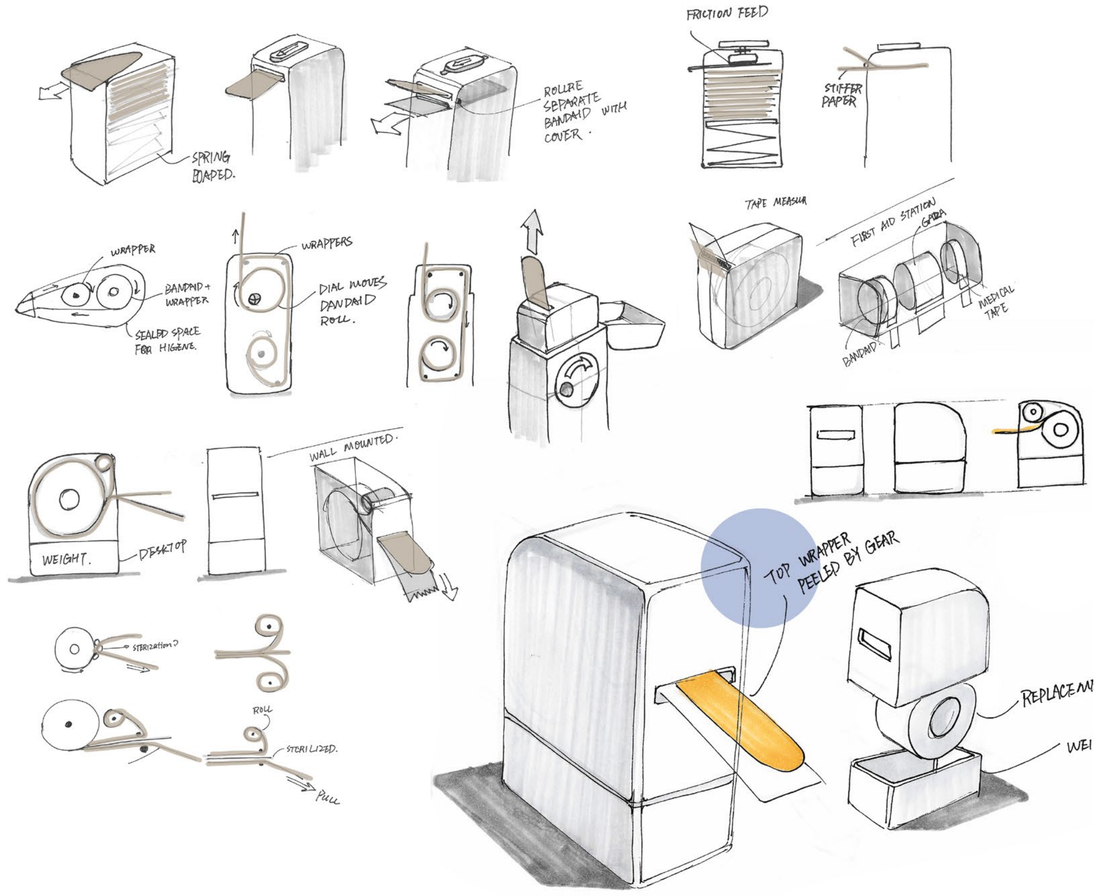


How might we simplify minor wound care, making bandage easier to access, prepare, and apply with one hand?

IDEATION

Ideation explored different ways to **dispense** and **unwrap** a bandage, along with **packaging** formats that could enable easier one-handed use.



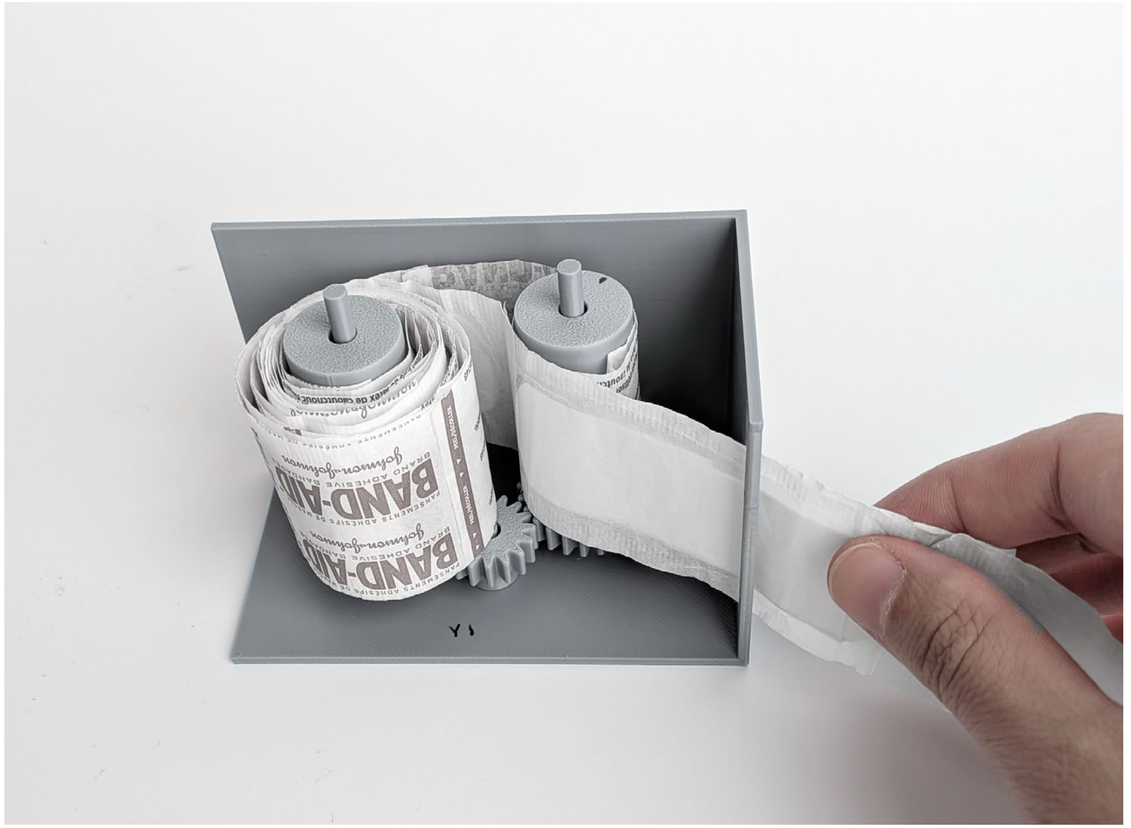


CONCEPT DEVELOPMENT

01 SPRING BOX

Wrapper adhesion created too much resistance, making the bandage **difficult to pull** and unwrap.

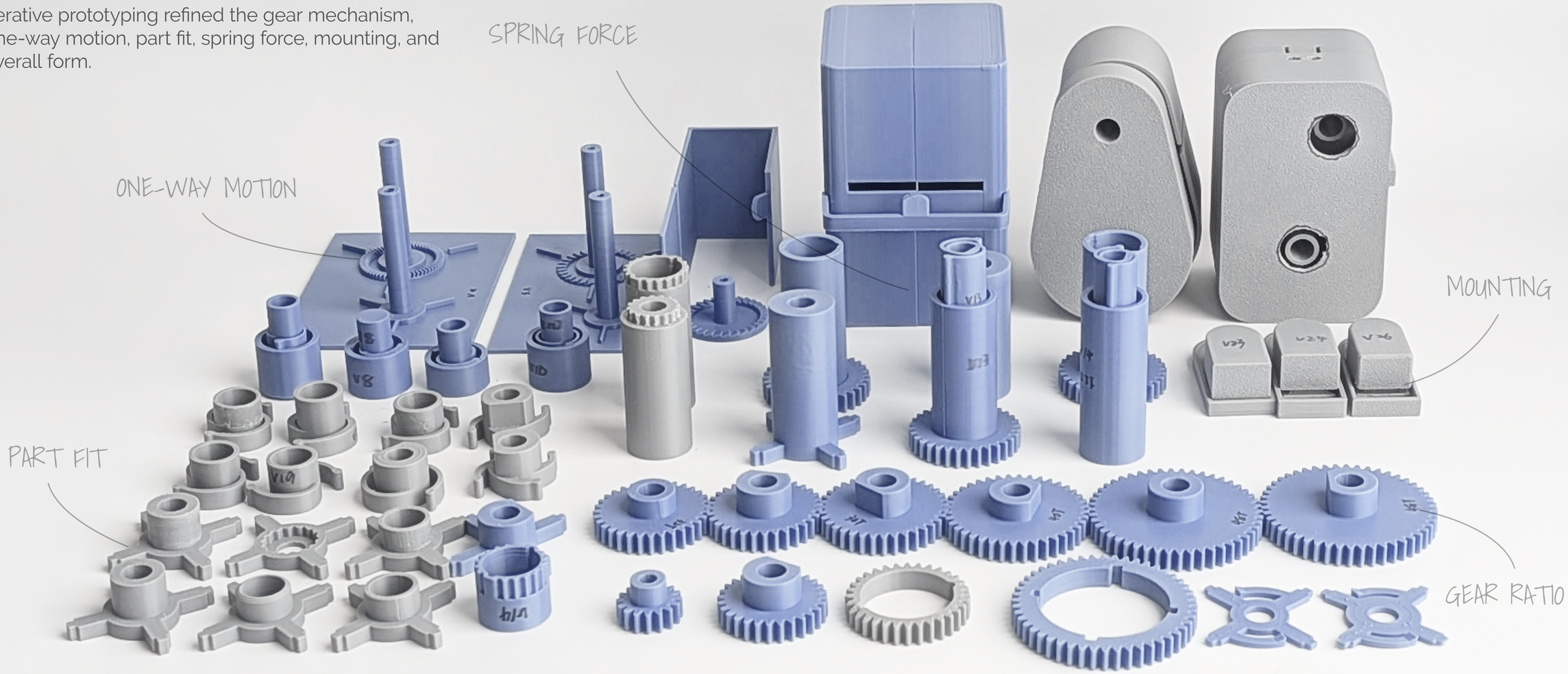
02 GEAR BOX



Gears assist the unwrapping motion with **less pull force**, showing more potential for further development.

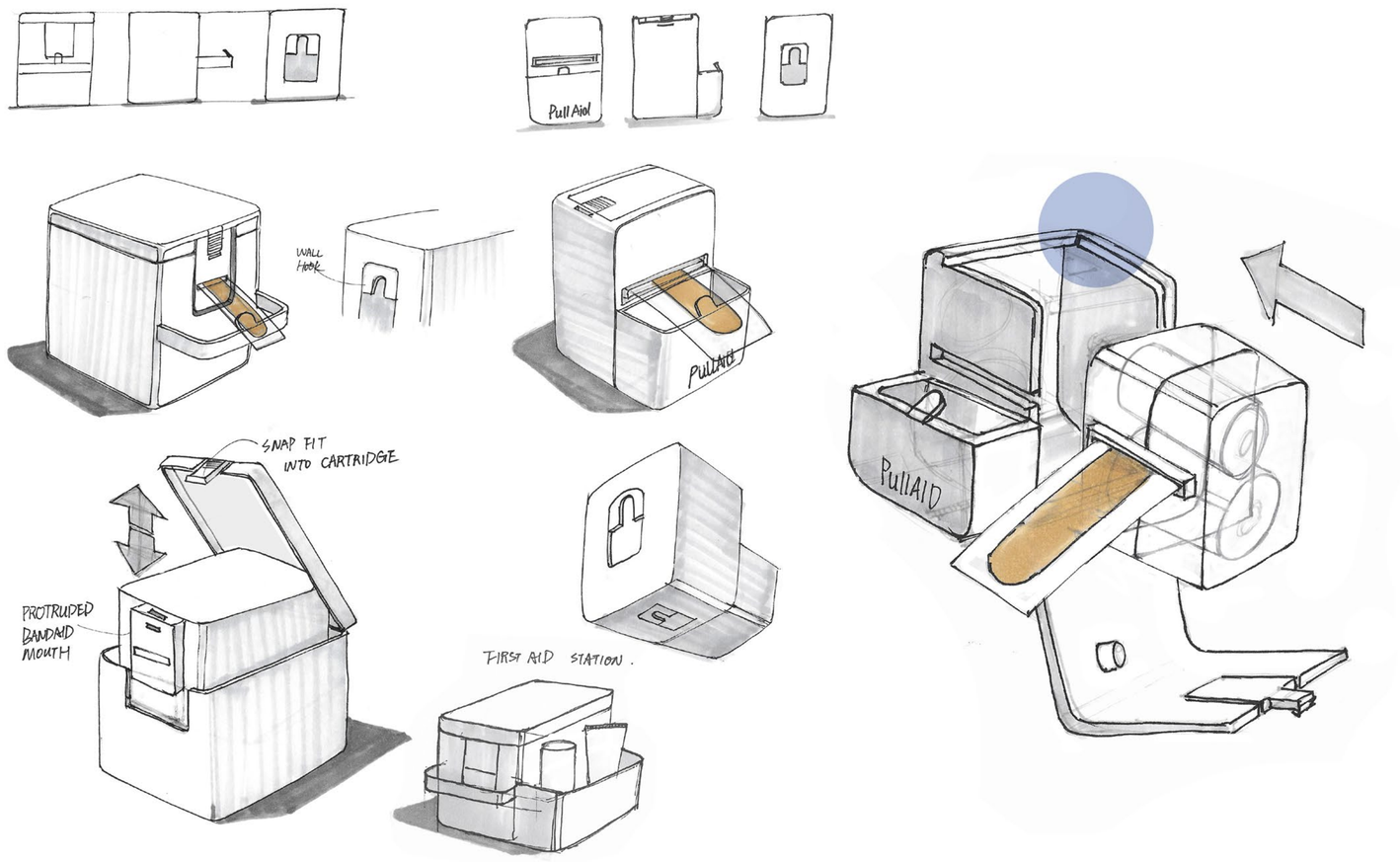
CONCEPT DEVELOPMENT

Iterative prototyping refined the gear mechanism, one-way motion, part fit, spring force, mounting, and overall form.



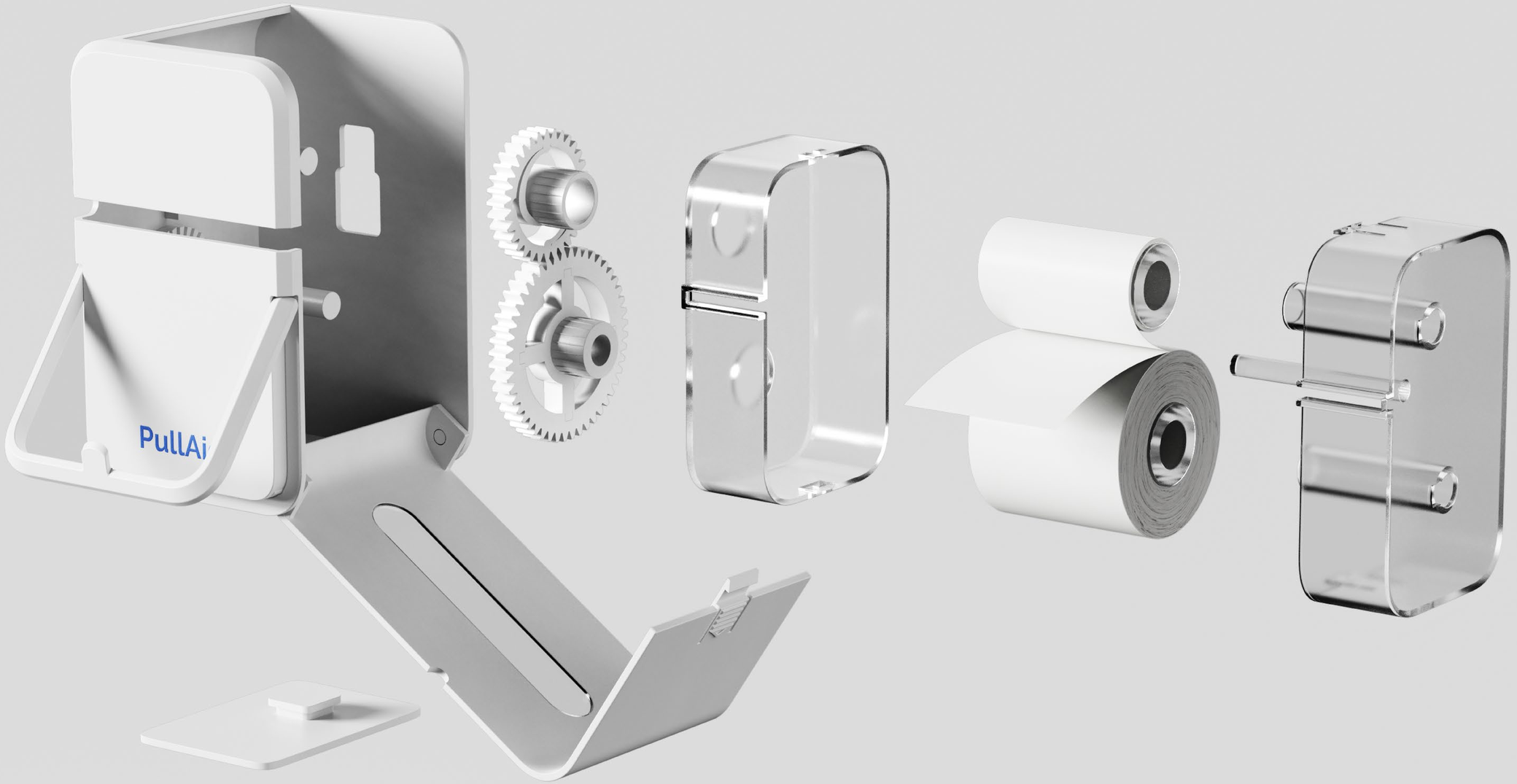
CARTRIDGE DEVELOPMENT

A cartridge system keeps the complex **bandage roll preloaded** for easy installation, while **reusing** the main dispenser to reduce waste.



PullAid

Pull. Peel. Protect.


PullAi

PullAid

THANKS

Let's Chat

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